

Ricardo (Zhicong) Xie

Building Services / HVAC Intern | Revit MEP, MVHR/ERV, CAD and Commissioning

Auckland, New Zealand | +64 29 027 29390 | zxie588@aucklanduni.ac.nz

LinkedIn: <https://www.linkedin.com/in/ricardo-xie-6394643b4> | Portfolio: <https://zxie588-alt.github.io> | GitHub: <https://github.com/zxie588-alt>

Valid NZ student visa | No sponsorship required | 25h/week during semester | Full-time 17 Nov 2026 - late Feb 2027 | Open to NZ-wide roles

PROFILE

- Auckland-based mechanical master's student with a NZ residential MVHR/ERV Revit MEP project.
- Built layout, airflow schedule, duct velocity checks, pressure estimate, equipment note and commissioning plan.
- Project screenshots and source links: <https://zxie588-alt.github.io>

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Relevant coursework: Building Services Engineering, Thermal Modelling, HVAC System Design, Engineering Methods

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Engineering team leadership: led project planning, task split and final technical presentation in team design work.
- Workshop practice: prepared AutoCAD/DXF laser-cut geometry and checked cut parts against drawings.
- Hands-on exposure: CNC machining, workholding, cutting fluid, chip formation and 3D printing workflow.
- Lab practice: interpreted force-displacement tests and controlled a robotic arm in training sessions.
- Technical communication: former student news lead; strong project records, photography and presentation visuals.

PROJECT WORK

Small Residential MVHR / ERV Revit MEP Coordination

Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes

- Built a NZ residential MVHR/ERV package with Revit MEP, CAD evidence and review notes.
- Modelled MVHR unit, ducts, terminals and hangers in SolidWorks; exported STEP, STL, DXF and PNG.
- Created a Revit 2026 model with plan, 3D view, airflow schedule, sheets and simple clash notes.
- Checked 60 L/s supply, 45 L/s extract and 85 L/s boost against duct velocity and pressure allowances.
- Added M-102 pressure review, BIM issue log and Mitsubishi Lossnay LGH-35RVX3-E selection note.
- Linked the design to NZBC G4/H1, Healthy Homes context and a commissioning test plan.

Thermal-Fluid Simulation & Validation - Solar Chimney Case Study

ANSYS Fluent/CFX, conjugate heat transfer, natural convection

- Built a CFD validation workflow for a laboratory-scale solar chimney with coupled air, absorber and cover domains.
- Checked mesh sensitivity, mass/heat balance, Reynolds/Rayleigh scaling and solver behaviour.
- Documented model boundaries, limitations and recommended physical data points needed for future experimental correlation.

Team Systems Engineering Feasibility Study

Stakeholders, requirements, ConOps, trade studies, risk, verification

- Led systems-engineering structure for a 5-person technical feasibility study.
- Built stakeholder map, requirements hierarchy, ConOps, trade criteria and risk register.
- Used verification gates to keep orbit, power, thermal, communications and cost assumptions aligned.

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared AutoCAD/DXF geometry for laser cutting and checked the cut plate against drawing intent.
- Practised CNC workflow: workholding, tool motion, cutting fluid, chip formation and shop safety.
- Reviewed machined gears and shafts for tolerance, edge quality and tool-access constraints.

Engineering Practice and Automotive Structure Training

Mechanics testing, robotic arm control, automotive structure walkthrough

- Worked with mechanics testing equipment and interpreted force-displacement results from a physical test setup.
- Practised robotic arm control and basic automation interaction in a lab setting.
- Led a small automotive-structure walkthrough covering seat/interior systems and vehicle context.

TECHNICAL SKILLS

- Revit MEP: MVHR/ERV layout, coordination plan, 3D view, airflow schedule and sheet setup.
- HVAC checks: airflow schedule, duct velocity, pressure-loss estimate and fan-duty assumptions.
- CAD support: SolidWorks, AutoCAD/DXF, component lists and colour-coded coordination layouts.
- NZ context: NZBC G4/H1 awareness, Healthy Homes ventilation and commissioning notes.
- Analysis support: heat transfer, natural-convection CFD and sensor-validation workflow.